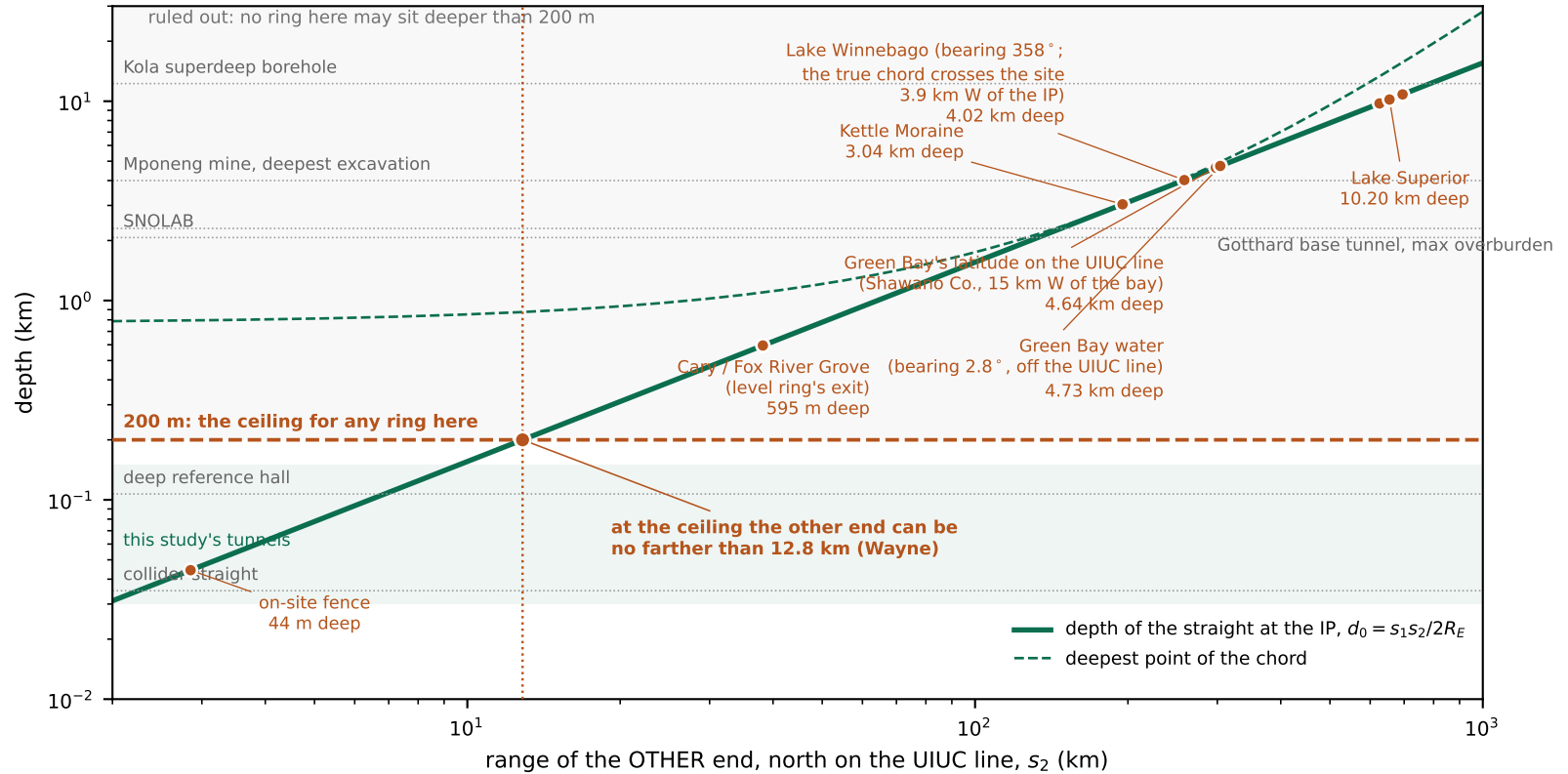


One straight with BOTH ends far: south end at UIUC (198 km), north end at s_2
the straight must sit $d_0 = s_1 s_2 / 2R_E$ below the IP – kilometres, for any northern water



A straight is a chord: its two exits obey $s_{near} s_{far} = 2R_E d_0$. With no ring deeper than 200 m, the UIUC straight's other end can reach 12.8 km at most; Green Bay would need 4.7 km, Lake Superior 10 km – deeper than any mine, drawn for scale. The feasible route is the ring's two straights aimed separately at ordinary depth (see text): inclined arcs and ~142 mrad of vertical bending per lap.